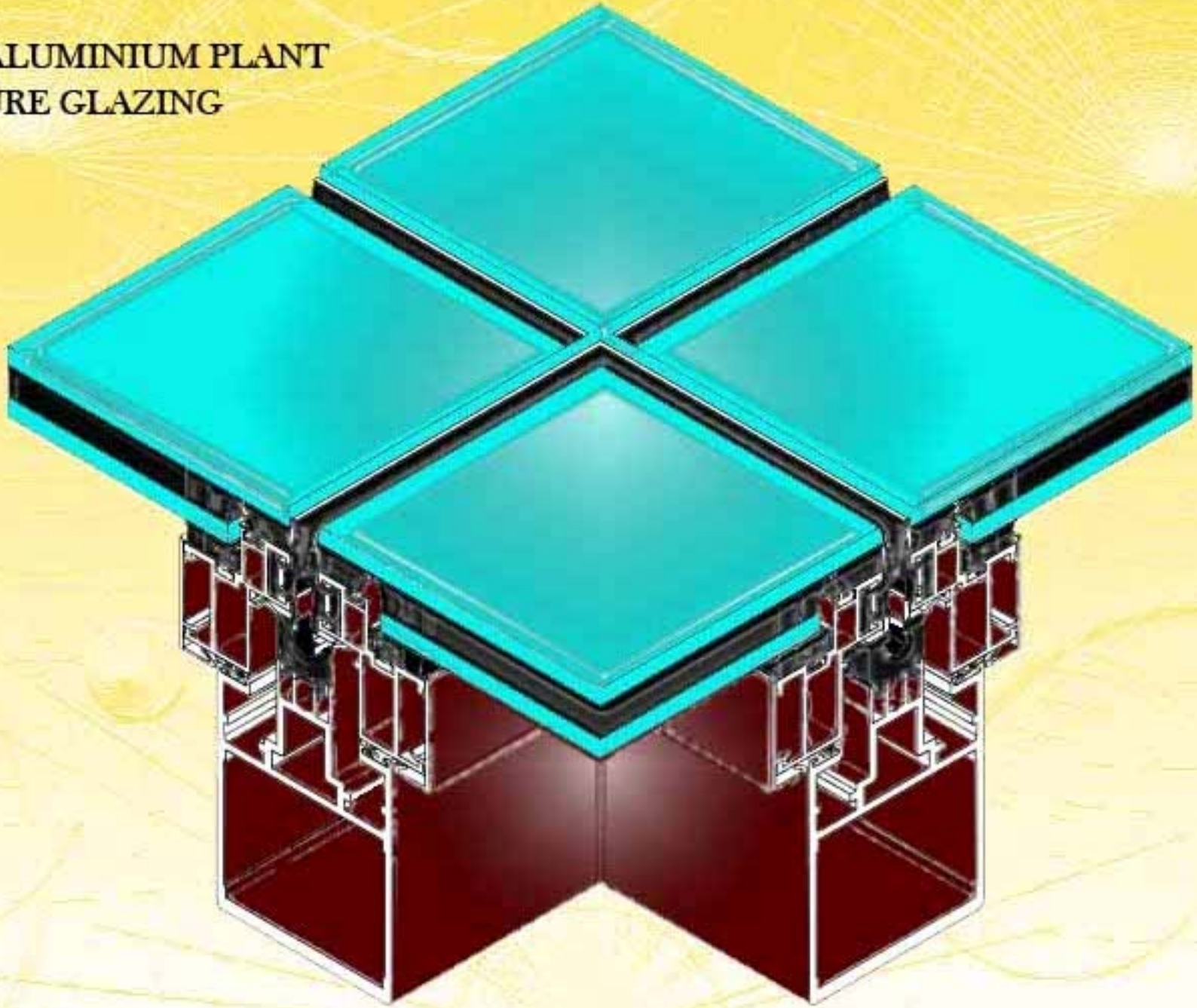
Structural Glazing
Curtain Wall System

Curtain Wall Systems

AL - JAZIRAH ALUMINIUM PLANT
STRUCTURE GLAZING



DH

مصنع ألمنيوم الجزيرة

Al Jazirah Aluminium Plant

Structural Glazing :

1 - A unique design in the Kingdom .
2 - A mixture of grace and quality .
3 - Complete sealing because of the presence of no less than five sealant points .
4 - Appearance of gap between glass panels lesser than the real .
5 - Special mastic sealant for holding the glasses for the strength & safety .
6 - Mastic sealant in three separate layers for a firm adhesion .
7 - The glass covered on the edges from all sides for beauty and for safety .
8 - Possibility of further fastening of glass from inside .
9 - European original made accessories .
10 - Alum. wall thickness at least 3 mm for solidity & quality.
11 - Screws invisible from inside or outside for beauty & looks.
12 - Depth of vertical aluminum up to 200 mm depend on height.
13 - Possibility of operable windows for this special system.

الواجهات الزجاجية :

1 - تصميم فريد في المملكة .
2 - إمتزاج الجمال و الجودة .
3 - منع التسرب الكامل بسبب وجود 5 ربلات كمصدات للتسربات .
4 - ظهور فراغ الزجاج بين لوحتين أقل من الواقع .
5 - تميز المعجون الخاص لإمساك الزجاج بالقوة والامان .
6 - تميز المعجون اللاصق في ثلاث درجات للتثبيت القوي .
7 - الزجاج مغطى من كل طرف للجمال و الوقاية الكاملة .
8 - إمكانية تشديد الزجاج إلى الداخل لمزيد من الربط .
9 - مزودة باكسسوارات اوروبية اصلية الصنع .
10 - وجود سماكات الألمنيوم العاليةالتي تبدأ من 3 مم للمتانة والجودة .
11 - تميز المقطع بالبراغي المخفية من الخارج والداخل للجمال والاناقة .
12 - توفر اعماق مختلفة من الألمنيوم العمودي إلى عمق 200 مم حسب الارتفاعات.
13 - إمكانية صنع النوافذ المتحركة لهذا النظام .

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STRUCTURAL GLAZING CURTAIN WALL SYSTEM
AJ – S.G.CW.

1- GENERAL

Description:

1.0 Structural type curtain wall:

The curtain wall shall be structural type AJ-S.G.CW. - from AL-JAZIRAH. Self supporting members shall have an overall depth of 100 mm by 170 mm site line with inner supporting framework made of aluminum alloy extruded sections and preassembled, pre-glazed outer panels placed on the framework with a visible gap (recess) between panels not exceeding 17 mm.

1.1

The framework must be made up of vertical mullions fixed to the slabs by means of brackets with adjustment on the three orthogonal directions and fixed with stainless steel bolts. For the structural calculations it is necessary to take into consideration the following loads: dead weight, wind action, thermal actions.

1.2 Structural see-through glaze panels:

The panels for the see-through areas shall be made up of an aluminum alloy extruded profile framework and an insulating glazing (or single glazing) which shall be applied through structural specifications. Such panels shall be assembled simultaneously to the construction of the insulating glazing.

1.2.1

If the see-through panels are to be operable, they should be of the top hinged project-out type, with balancing arms, handle, and lock gear bolt (see hardware list).

1.3 Glazing:

The glazing shall be top quality, float tinted or reflective type and without any manufacturing defects according to International Standard, with the following characteristics:

Glass thickness: as shown in window type.....

Glass color: as described.....

U Value (ISO) of (varies per glass type) W/m2. K

Solar energy reflection: .....%

Solar energy transmission: ...%

Daylight reflection: .....%

Daylight transmission: .....%

1.4.0 Mechanical, insulating and sealing features:

The curtain wall shall correspond to the characteristics of Static Calculation, Thermal Expansion, Tolerances, Thermal Insulation of the area and orientation of the building, and Condensation Resistance Factor (C.R.F). All work shall be carried out according to ASTM 1503.1-88 and ES1787/89.

The following calculations shall be furnished:

- Structure resistance
- Glass resistance

1.5.0 Quality Assurance:

Drawings and specifications are based upon the structural curtain wall series four side structural silicone manufactured and installed by the AL-JAZIRAH Plant, supported by technical literature samples, shop drawings & performance data.

1.6.0 Performance Requirements:

- Air infiltration shall be tested in accordance with ASTM E-283-84. Infiltration shall not exceed 0.06 CFM per square foot (.00005 M3/SM2) of frame area.

- Water infiltration shall be tested in accordance with ASTM E-331-86 for uniform static air pressure and ASTM E-547- 86 for cyclic air pressure differential No water penetration at a test pressure of 10 PSF.

- De-glazing force shall be tested in accordance with ASTM E-987-88; Force to deglaze panels shall be not less than 155 lb. (70 Kg) on vertical and horizontal members.

- Forced entry resistance shall be tested in accordance with AAMA1303.5-76.

- Structural performance shall be tested in accordance with ASTM E-330-84 and AAMA 101-88. When tested at the design pressures defined below applied for a period of 10 seconds, there shall be no permanent deformation of any frame or panel member also, there shall be no glass breakage, permanent damage to fasteners, hardware parts, actuating mechanisms, or other damage, which would cause the concealed windows to be inoperable. HC-45 rating of 45 PSF (1.90 Kg/M2) positive and negative design load, with deflection not to exceed L/175 of its clear span.

II. PRODUCTS

II.1.0 Materials, finishing and qualities:

II.1.1 Materials:

Extrusions shall be 6063-T6 alloy and temper (ASTM B-221 alloy G.S lOA-T6). Fasteners, where exposed, shall be aluminum, stainless steel or plated steel in accordance with ASTM A-164, Perimeter anchors shall be aluminum or steel, providing the steel is properly isolated from the aluminum Glazing gaskets shall be EPDM extrusions.

II.2.0 Finishing:

II.2.1

Powder coating for all aluminum parts shall be made of Pure Polyester Powder with film thickness 60 to 90 microns. Appropriate 8 stages, chromate pretreatment should be carried out before applying powder coat. (See attached specifications for powder coating).

11.2.2

Anodizing finish is obtained by electrolytic impregnating tin ions into the oxide film to a depth of 15 – 18 microns , the colors obtained can be gold , silver or bronze color .

II.3.0 Qualities:

II.3.1

All steel products shall be hot galvanized.

II.3.2

All screws and bolts shall be made of stainless steel and /or chromate steel.

II.3.3

All the weather stripping and/or finishing gaskets shall be made of EPDM.

II.3.4

All the sealant used shall be compatible with the various uses according the international standers

II.3.5

All the coupling and/or contacts between the aluminum components and other materials shall be protected.

Rubber Gaskets:

Tightness of all operable sashes (hinged), and of glazing is assured by EPDM gaskets. EPDM (Ethylene Propylene Dine Monomer) is a rubber mixture extremely resistant to severe weather changes. Acids do not affect it, alkalinize, Ozone, salts, heat, cold, nor moisture.

Specifications of rubber gaskets can be as follows:

- The gaskets' material shall consist of at least 50% by weight of basic rubber hydrocarbon.
- The material shall not contain any crude or reclaimed rubber.
- The material shall be homogeneous, and free from defects.
- Cured EPDM shall meet the following properties when tested in accordance with ASTM test slabs and compression set plugs:

Property	**Test Procedure**	**Typical Values**
Hardness (±5)	ASTM D - 2000	50
Tensile Strength (N/ sq. mm)	ASTM D - 2000	11.2 ® 12.4
Elongation (%)	ASTM D - 412	400
Heat Aging	ASTM D - 865	Complies

II.3.6 Hardware:

(1) Set of heavy-duty four bars adjustable balancing arms to enable the smooth. Movements of large sashes (up to 120Kgm) for single sash out swing.

(1) Locking gear bolt set.

1 lock set with (6) six points locking plates.

1 fixed handle (removable handle option).

1 set of limit stop out swing (option).

III. EXECUTION

III.1.0 Installation:

All jambs, head and sill track shall be set in correct locations as shown in the details and shall be level, square, plumb and in alignment with other work in accordance with the manufacturer's installation instructions and. approved shop drawings. All joints between framing and the building structure shall be sealed in order to secure a watertight installation.

III.2.0 Protection & Cleaning:

After installation, the General Contractor shall adequately protect exposed portions of aluminum surfaces from damage by grinding and polishing compounds, plaster, lime, acid, cement, or other contaminants.

The General Contractor shall be responsible for final cleaning.